Sahand Rezaei-Shoshtari

PhD Candidate in Computer Science

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Education

Present – Sep. 2020 | **PhD, School of Computer Science, McGill University and Mila, Montreal, Canada**
Supervisors: Doina Precup, David Meger
CGPA: 4.00/4.00

Dec. 2019 – Sep. 2017 | **Master of Engineering - Thesis, McGill University, Montreal, Canada**
Supervisors: Inna Sharf, David Meger
CGPA: 4.00/4.00
Thesis: Learning Manipulator Dynamics for Control and Interaction Inference

Sep. 2016 – Sep. 2012 | **Bachelor of Mechanical Engineering, University of Tehran, Tehran, Iran**
CGPA: 3.98/4.00
Thesis: Online Path Planning for a Mobile Robot in Dynamic Environments using Reinforcement Learning

Work Experience

Present – Apr. 2024 | **Researcher, Microsoft Research AI for Science, Montreal, Canada**
› AI for science and drug discovery

Aug. 2023 – May. 2023 | **Research Intern, Microsoft Research AI for Science, Amsterdam, The Netherlands**
› Deep reinforcement learning for chemical reactions discovery

Feb. 2023 – Apr. 2022 | **Research Intern, Samsung AI Center, Montreal, Canada**
› Hypernetworks for meta imitation learning and meta reinforcement learning

Sep. 2020 – Mar. 2020 | **Research Intern, Samsung AI Center, Montreal, Canada**
› Multimodal generative modeling for learning intuitive physics
› Deep reinforcement learning for load balancing of 5G networks

Mar. 2020 – Jan. 2020 | **AI Programmer, Ubisoft La Forge, Montreal, Canada**
› Deep reinforcement learning for automated video game testing

Aug. 2019 – Mar. 2019 | **Research Intern, Samsung AI Center, Montreal, Canada**
› Object detection neural networks for human-robot interaction

Publications

2024 **Rezaei-Shoshtari, S.**, Fujimoto, S., Yurchyk, H., Precup, D., and Meger, D., 2024. "Long-Term Fairness in Reinforcement Learning with Bisimulation Metrics". *Under Review*.

2024 Panangaden, P.*, **Rezaei-Shoshtari, S.***, Zhao, R.*, Meger, D., and Precup, D., 2024. "Policy Gradient Methods in the Presence of Symmetries and State Abstractions". *Journal of Machine Learning Research (JMLR)*.

2023 **Rezaei-Shoshtari, S.**, Morissette, C., Hogan, F.R., Dudek, G., and Meger, D., 2023. "Hypernetworks for Zero-shot Transfer in Reinforcement Learning". *The AAAI Conference on Artificial Intelligence*.

2022 **Rezaei-Shoshtari, S.**, Zhao, R., Panangaden, P., Meger, D., and Precup, D., 2022. "Continuous MDP Homomorphisms and Homomorphic Policy Gradient". *Advances in Neural Information Processing Systems (NeurIPS)*.

2021 **Rezaei-Shoshtari, S.**, Hogan, F.R., Jenkin, M., Meger, D., and Dudek, G., 2021. "Learning Intuitive Physics with Multimodal Generative Models". *The AAAI Conference on Artificial Intelligence*.

2021 Hogan, F.R., Jenkin, M., **Rezaei-Shoshtari, S.**, Girdhar, Y., Meger, D., and Dudek, G., 2021. "Seeing Through your Skin: Recognizing Objects with a Novel Visuotactile Sensor". *The IEEE/CVF Winter Conference on Applications of Computer Vision*.

2020 Molamohammadi, M., **Rezaei-Shoshtari, S.**, and Quitoriano, N., 2020. "Jacobian of generative models for sensitivity analysis of photovoltaic device processes". *The Machine Learning for Engineering Workshop at NeurIPS*.

2020 **Rezaei-Shoshtari, S.**, Meger, D., and Sharf, I., 2020. "Learning the Latent Space of Robot Dynamics for Cutting Interaction Inference". *The IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.

2019 **Rezaei-Shoshtari, S.**, Meger, D., and Sharf, I., 2019. "Cascaded gaussian processes for data-efficient robot dynamics learning". *The IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.

Skills

Programming	Python, C++, C#, MATLAB
Machine Learning Frameworks	PyTorch, TensorFlow, Hugging Face Transformers, LangChain, GPyTorch
Platforms	Docker, Slurm Workload Manager, ROS
Other Software	Git, Mujoco, Unity 3D, OpenCV, PyBullet, MoveIt

Select Projects

Hypernetworks for Zero-Shot Transfer in Reinforcement Learning — 2023

Paper Webpage github.com/SAIC-MONTREAL/hyperzero

A novel algorithm based on hypernetworks for zero-shot transfer in deep RL.

Reinforcement Learning | Hypernetworks | Transfer Learning | PyTorch | Python

Contextual Control Suite — 2023

Paper github.com/SAIC-MONTREAL/contextual-dm-control

A series of contextual Markov Decision Processes (MDP) based on Mujoco for continuous control tasks.

Reinforcement Learning | Mujoco | Gym | Python

Deep Homomorphic Policy Gradient (DHPG) Algorithm — 2022

Paper Webpage github.com/sahandrez/homomorphic_policy_gradient

A novel algorithm for state-action abstraction and representation learning in deep RL.

Reinforcement Learning | Representation Learning | PyTorch | Python

Multimodal Generative Modeling and Visuotactile Simulation — 2020

Paper Webpage github.com/SAIC-MONTREAL/multimodal-dynamics

A novel algorithm for multimodal generative modeling of intuitive physics.

Generative Modeling | Multimodal | PyTorch | PyBullet | Python

Gym Forest Fire — 2020

github.com/sahandrez/gym_forestfire

A Gym environment of a forest fire simulation for tackling wildfires with RL.

Reinforcement Learning | Gym | Python

Control and Motion Planning Utilities for Kinova Jaco 2 Robot — 2019

github.com/sahandrez/jaco_control

A ROS package for the Kinova Jaco 2 robot with various control and motion planning utilities.

Robotics | ROS | Python

Select Honors and Awards

2022 – 2025	Canada Graduate Scholarship-Doctoral (CGS-D) Award. Amount of **$105,000**. NSERC.
2022 – 2026	Fonds de Recherche du Quebec - Nature et Technologies (FRQ-NT) Award. Amount of **$70,000**. FRQ.
2022	Grad Excellence Award. Amount of **$7,000** per year. McGill University.
2022	AAAI Student Scholarship Program. AAAI 2023.
2020	DeepMind Grad Award. Amount of **$25,000**. DeepMind and McGill University.
2019	IROS Student and Developing Countries (SDC) Travel Award. IEEE/RSJ IROS 2019.
2017	Grad Excellence Award. Amount of **$5,000**. McGill University.
2012	National University Entrance Exam. Ranked **19th**. Iran.

Community Involvement

2024 – Present	Senior reviewer at RLC.
2022 – Present	Reviewer at NeurIPS, ICML, ICLR.
2022	NeurIPS 2022 Outstanding Reviewer (Top 8%). NeurIPS.
2022	ICML 2022 Outstanding Reviewer (Top 10%). ICML.

Certifications

Apr. 2023	**Trustworthy and Responsible AI Learning (TRAIL)**, Mila, Montreal, Canada
Oct. 2020	**Simons Institute Mathematics of Online Decision Making Workshop**, Virtual
Aug. 2019	**CIFAR Deep Learning and Reinforcement Learning (DLRL) Summer School**, Edmonton, Canada